

JOHN KAMAU

FULL-STACK SOFTWARE ENGINEER & SYSTEMS ARCHITECT

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with dedicated expertise in distributed backend systems, event-driven architectures, and responsive modern web applications. Proven track record building production-grade IoT telemetry platforms, e-commerce marketplaces with Safaricom Daraja M-Pesa automated payouts, and resilient microservices.

TECHNICAL CAPABILITIES

Languages:	Python, TypeScript, JavaScript, Go (Golang), SQL, Bash
Frameworks & Runtimes:	Next.js (App Router), React, FastAPI, Django, Node.js, Express, Tailwind CSS
Databases & Cache:	PostgreSQL, TimescaleDB, Redis, SQLite
Cloud & Infrastructure:	Docker, Linux / Unix, Git & GitHub Actions, RESTful APIs, GraphQL
Specialized Integrations:	Safaricom Daraja 2.0 (M-Pesa STK Push), IoT Telemetry, Webhooks, Microservices

FLAGSHIP ENGINEERING PROJECTS

Smart Poultry — Comprehensive IoT Telemetry & Farm Management Platform

Python • Django / FastAPI • PostgreSQL • Edge IoT Sensor Telemetry

- Engineered an automated farm telemetry system tracking real-time brooder ambient temperature, humidity, and feed consumption.
- Integrated threshold alerting via push notifications and SMS, cutting mortality rates during brooding cycles by 35%.
- Architected historical sensor telemetry storage and analytics dashboards for flock growth curve modeling.

Market Mtaani — Rural Business Hub & Direct Artisan Marketplace

Python • Django • Safaricom Daraja 2.0 M-Pesa STK • PostgreSQL

- Built a digital exchange linking smallholder farmers and rural artisans directly with regional buyers, cutting out middlemen.
- Engineered automated mobile escrow checkout using Safaricom Daraja 2.0 M-Pesa STK push with idempotent webhook verification.
- Designed mobile-responsive web storefronts optimized for 3G mobile devices with low-bandwidth image payloads.

Django Enterprise ERP & Core Architecture Modules (djangosystem)

Python • Django REST Framework • PostgreSQL • Docker

- Developed a modular ERP core supporting role-based access control (RBAC), multi-tenant schemas, and audit logs.
- Optimized database indexes and query prefetching, achieving sub-40ms response times on high-concurrency API endpoints.
- Packaged complete CI/CD pipelines with automated unit testing and containerized Docker deployments.

High-Performance URL Shortener & Analytics Gateway

Python • FastAPI • Redis • REST API

- Designed a high-throughput URL hashing and redirection service with in-memory Redis caching.
- Integrated real-time click analytics, geographic IP breakdown, and rate limiting per client API token.

EDUCATION & CONTINUOUS LEARNING

Software Engineering & Distributed Systems Development

Focus on Backend Architecture, Data Modeling, API Design, and Cloud Native Deployments.

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